

Reward Shaping and Curriculum PPO for Sparse-Reward SoccerTwos Control

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Abstract: Our project studies PPO-based policy learning in SoccerTwos, a sparse-reward Unity soccer environment where agents must learn to reach the ball, move it towards goal and score using limited feedback. We examined two changes to a basic PPO baseline that was learning slowly. Firstly, curriculum learning with progressively harder start states. Then, reward shaping with small, dense bonuses for ball-directed progress. Curriculum training yielded the strongest policy, while reward shaping enhanced early training from 0.2399 to 0.4036 mean episode reward at an identical 2M-step budget. After about 30 million environment steps, the final curriculum v3 agent, exported as `TEAMNAME_v3_AGENT`, achieved a final mean episode reward of 1.9692. It also won eight out of ten recorded matches against the PPO baseline. These findings imply that curriculum learning was the best approach to overcome the primary bottleneck, which was sparse-reward exploration.

1 Overview

In the 2-team Unity soccer game SoccerTwos, cube agents must interact to learn how to move, approach the ball, defend and score goals. The default reward is sparse making the task difficult because until the agent achieves significant soccer events like scoring or conceding, many trajectories include little helpful feedback. Because PPO is stable and popular for continuous interaction contexts, and supported by the RLlib training pipeline implemented in the project, we approached this as a policy-learning issue and chose it as the basis method. The primary objective was find out if PPO might learn better soccer behavior with more structure. Because of this, we compared a sparse-reward PPO baseline, a reward-shaped PPO agent with dense progress signals, and a curriculum PPO agent that starts from easier states before training on harder game situations. The final submitted policy, `TEAMNAME_v3_AGENT`, was given by extending the best curriculum PPO checkpoint to a longer training budget.

2 Method Description

PPO baseline. The baseline used RLlib PPO with the default sparse SoccerTwos reward and a feed-forward ReLU actor-critic network. PPO was chosen because its clipped objective limits destructive policy updates from rollout data. The baseline and reward-shaped agents used a shared policy/value model with one 512-unit hidden layer.

Reward-shaped PPO. To reduce sparse-reward exploration difficulty, the shaped agent added a small training-only reward from the environment info dictionary. Let d_{pb} be player-to-ball distance and d_{bg} be ball-to-goal distance. At each transition,

$$r_{\text{shape}} = \text{clip} \left(0.01(d_{pb}^{t-1} - d_{pb}^t) + 0.02(d_{bg}^{t-1} - d_{bg}^t), -0.05, 0.05 \right).$$

This rewards moving toward the ball and moving the ball toward the opponent goal, while clipping keeps the bonus small relative to scoring rewards. The exported policy does not use privileged

Table 1: Main PPO training configurations used in the project.

Method	Reward	Curriculum	Network
PPO baseline	sparse	no	[512]
Reward-shaped PPO	sparse + shaping	no	[512]
Curriculum PPO	sparse + curriculum setup	yes	[256, 256]
Curriculum PPO v3	resumed curriculum	yes	[256, 256]

info fields at inference time. This follows the usual motivation for reward shaping in sparse-reward RL [1].

Curriculum PPO. The curriculum agent changed the training distribution by starting from easy game states and gradually increasing difficulty. A callback set initial ball and player states through the Unity environment channel and advanced through five tasks, from “Very Easy Goal” to “Random Players,” once `episode_reward_mean` exceeded 1.5. The curriculum policy used a shared-value ReLU MLP with hidden layers [256, 256]. The final v3 policy resumed the strongest curriculum checkpoint and trained for about 30.0M steps so the agent could stabilize on later randomized stages after learning easier early tasks [2].

Final hyperparameters. Our submitted policy was trained with Ray RLlib PPO using the PyTorch backend. In the final v3 run, we used a learning rate of 5×10^{-5} and discount factor $\gamma = 0.99$. We also used PPO clipping parameter 0.3, 30 SGD epochs per update, and train batch size 80,000. Others included minibatch size 8,000, rollout fragment length 2,000, gradient clipping at 0.5, 40 rollout workers, and 1 GPU. The exported submission package was `TEAMNAME_v3_AGENT`.

3 Preliminary Results

Figure 1 shows the main training comparison. Plain PPO struggled with the sparse reward, reaching only 0.2399 mean episode reward after 2.0M steps. Reward shaping improved this to 0.4036 at the same budget, but the largest gain came from curriculum learning. The first curriculum run reached 1.7869 mean reward after 2.06M steps, and the extended curriculum v3 run reached 1.9692 final mean reward, with a best logged value of 1.9705 at 29.85M steps. These curves suggest that curriculum design was more important than reward shaping alone for learning useful SoccerTwos behavior.

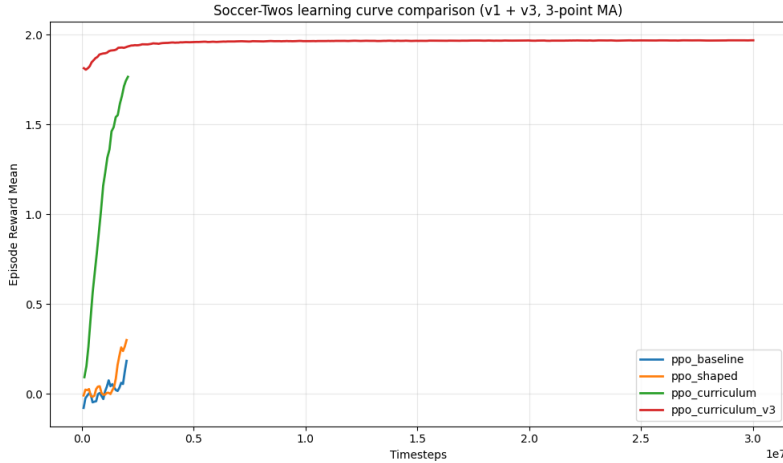


Figure 1: Mean episode reward over training for baseline PPO, reward-shaped PPO, curriculum PPO, and the extended curriculum v3 policy. Reward shaping improves over the sparse-reward baseline, but curriculum training gives the largest gain.

Table 2 reports the recorded headless evaluation matches for the final curriculum v3 policy. Because the number of evaluation episodes is small, these results should be treated as preliminary rather than

Table 2: Recorded headless evaluation summaries for the final curriculum v3 policy. Match counts are small, so these results are preliminary.

Matchup	Episodes	Wins	Agent mean	Opponent mean
Curriculum v3 vs. baseline	10	8	1.1140	-1.2084
Curriculum v3 vs. curriculum	10	6	0.3407	-0.4258

definitive. Still, the v3 policy won 8 of 10 matches against the PPO baseline and 6 of 10 matches against the earlier curriculum policy, which is consistent with the learning-curve improvement.

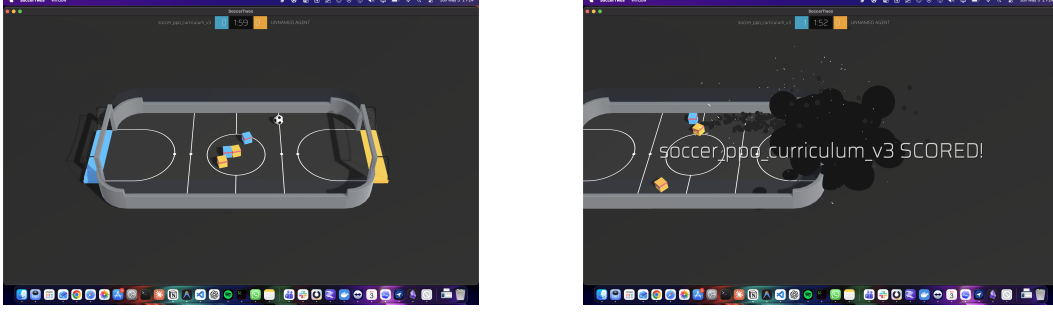


Figure 2: Gameplay screenshots from the final TEAMNAME_v3_AGENT evaluated in the SoccerTwos environment. The agent (blue team) consistently moves toward the ball and positions itself to score.

Figure 3 shows the detailed training diagnostics for the final curriculum v3 run. The mean episode reward quickly approached 2.0 and remained stable, while episode length dropped sharply and then stabilized. This suggests that the final policy learned to reach terminal soccer outcomes more efficiently. The PPO diagnostics also show decreasing entropy and value loss, which is consistent with the policy becoming less random and the critic fitting the return signal more accurately.

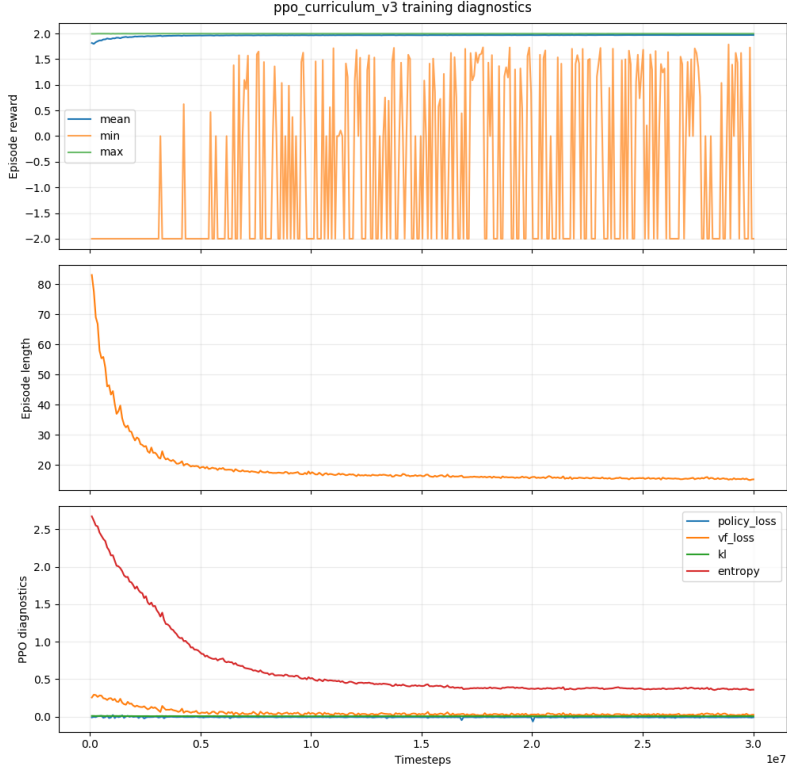


Figure 3: Detailed diagnostics for the final ppo_curriculum_v3 run. Mean reward stabilizes near 2.0, episode length decreases substantially, and PPO diagnostics show stable convergence behavior.

4 Reward, Observation, and Architecture Modifications

Reward modification. The main reward-level change is the training-only `RewardShapingWrapper`. It uses `player_info` and `ball_info` to reward progress toward the ball and ball progress toward the opponent goal. The shaping bonus is clipped to ± 0.05 and added only during training, so the exported policy does not require privileged `info` fields. This improved the 2.0M-step mean reward from 0.2399 for baseline PPO to 0.4036 for reward-shaped PPO, helping early exploration but not producing the strongest final policy.

Observation and action handling. No custom observation-space wrapper or Unity-side observation rewrite was used. The policy uses the provided 336-dimensional `SoccerTwos` observation vector. The main interface adaptation is action handling: training uses flattened `Discrete(27)` actions, while live matches use branched `MultiDiscrete([3,3,3])` actions. Therefore, this report does not claim an observation modification.

Architecture and curriculum modification. The baseline and reward-shaped PPO agents use one 512-unit hidden layer, while the curriculum PPO agents use a two-layer $[256, 256]$ ReLU MLP with shared value layers. Curriculum training also changes the initial-state distribution, starting with easier near-goal situations before progressing to harder ball and player placements. This was the most effective modification, producing much higher training rewards and better recorded match outcomes than baseline PPO.

5 Limitations and Future Work

The main limitation is evaluation size. The strongest recorded comparison against the PPO baseline used only 10 headless episodes rather than a larger 100-episode evaluation, so the match results should be treated as preliminary. Training was also done in a simplified single-player setting against a fixed opponent, so the final policy should not be interpreted as learning full multi-agent coordination such as passing or team positioning. Imitation learning was explored, but it was not part of the final successful method. Future work should run larger evaluations against random, baseline, and TA agents; compare reward-shaped PPO directly against curriculum v3; and test whether the curriculum policy remains robust under self-play or stronger multi-agent opponents.

6 GitHub Link and Acknowledgments

https://github.com/Vedaang-Chopra/RL_Soccer_project

We used ChatGPT [3] and Gemini [4] for feedback, editing suggestions, code editing, bug fixing, formatting, brainstorming and $\text{\LaTeX}$ cleanup. All experiments, implementation choices, reported results, and final claims are the authors’ original work.

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